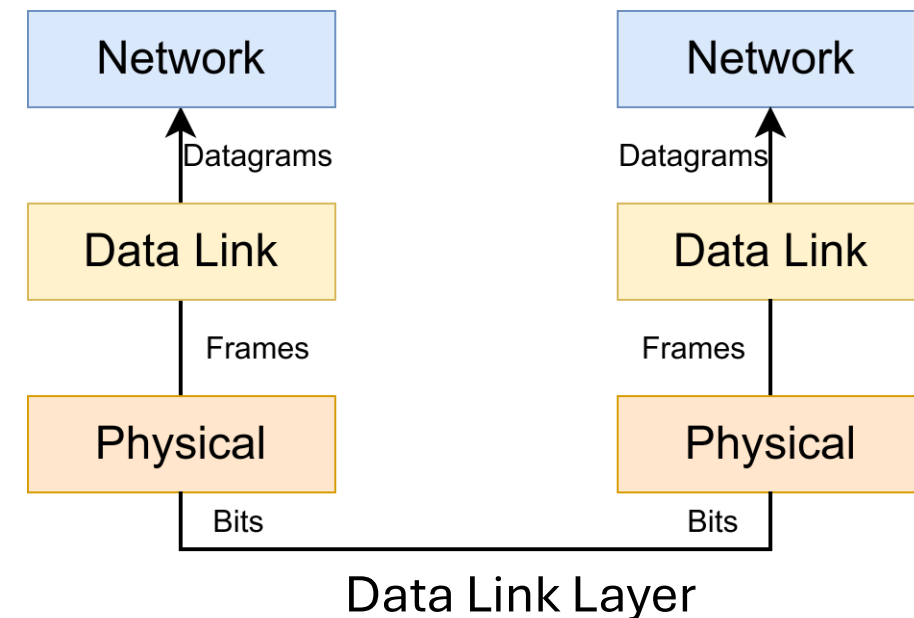


DATA LINK LAYER

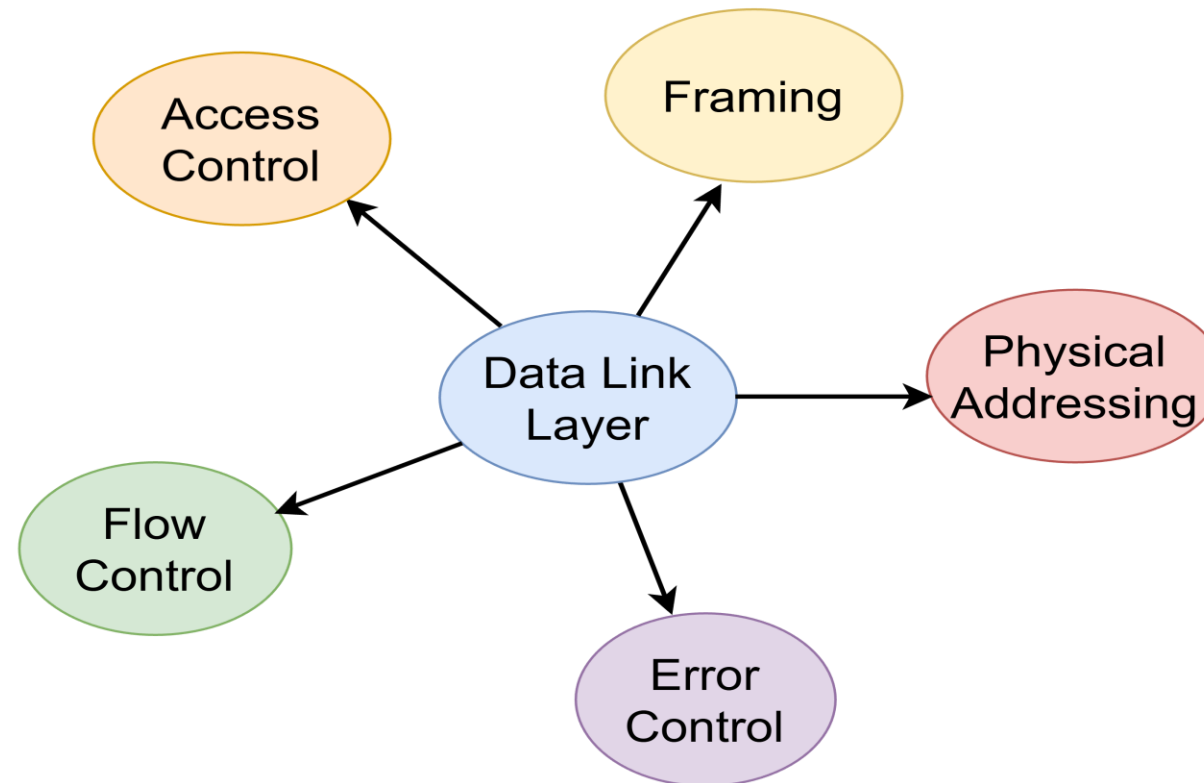
Module 2

Data Link Layer

- The duty scope of the data-link layer is **node-to-node**. When a **packet** is travelling in the Internet, the data-link layer of a node is responsible for **delivering a datagram** to the **next node** in the path.
- Sender **encapsulates the datagram** received from the network in a frame, and receiver needs to **decapsulate the datagram** from the frame.
- Manage the **flow of data** over physical medium.
- Responsible for **error-free transmission** over the **physical layer**.
- Assure **error-free data submission** to the **network layer**.



Functions of Data Link Layer



Functions of Data Link Layer

- **Framing:** Divides the **datagram** into **frames**.
- **Physical Addressing:** Adds the **sender** and **receiver physical addresses** to the **frame**.
- **Flow Control:** Controls the **speed of data transmission** so that the receiver can **receive data properly**.
- **Error Control:** **Detects errors** and **retransmits damaged** or **lost frames**.
- **Access Control:** Determines which device can **use the shared link** when **multiple devices share the same link**.

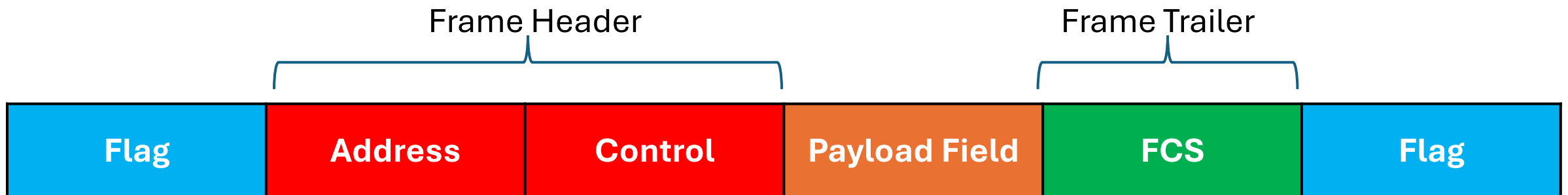
Data Link Layer Frame:

- A frame is a **basic unit** of communication in the data link layer.
- Data link layer takes the **packets** from the Network Layer and **encapsulates them** into **frames**.
- If the **frame size** becomes **too large**, then the **packet** may be divided into **small sized frames**.
- At **receiver's end**, data link layer picks up **signals** from hardware and **assembles** them into **frames**.

Fields of a Data Link Layer Frame:

A data link layer frame has the following parts:

- **Frame Header:** It contains the **source** and the **destination addresses** of the frame and the **control bytes**.
- **Payload field:** It contains the **message** to be delivered.
- **Trailer:** It contains the **error detection** and **error correction** bits. It is also called a **Frame Check Sequence (FCS)**.
- **Flag:** **Two flags** at the two ends mark the **beginning** and the **end** of the frame.



Frame Header:

A frame header contains the **destination address**, the **source address** and **three control fields** **kind**, **seq**, and **ack** serving the following purposes:

- **kind:** The kind field tells the receiver **what type of frame it is**, meaning that whether the frame is a **data frame** or used for **control functions** like error and flow control or link management etc. **Link management** refers to **establishing, maintaining, or terminating a link**.
- **seq:** This contains the **sequence number** of the frame for rearrangement of **out-of-sequence frames** and sending acknowledgments by the receiver. seq helps the receiver **identify and arrange** frames in their **correct order**.

Example:

Suppose A sends three frames: Frame 1, Frame 2, and Frame 3 with seq 0, seq 1, and seq 2, respectively.

Imagine the frames travel through a network and arrive in the **wrong order**:

Sent: Frame 1 $\rightarrow$ Frame 2 $\rightarrow$ Frame 3

Received: Frame 2 $\rightarrow$ Frame 1 $\rightarrow$ Frame 3

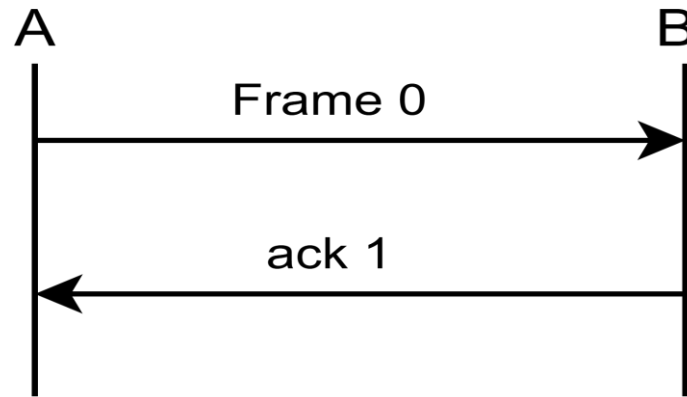
The receiver can look at the sequence numbers:

Received: Frame 2 $\rightarrow$ seq 1, Frame 1 $\rightarrow$ seq 0, Frame 3 $\rightarrow$ seq 2.

It can therefore understand that **Frame 1** should come **before Frame 2**.

- **ack:** This contains the **acknowledgment number** of some frame, particularly when piggybacking is used.

Example:



Here, B sends an acknowledgment to A.

In a common numbering convention, **ack 1** can mean that **Frame 0** has been **received**, and **Frame 1** is **expected next**.

This exact meaning depends on the protocol being used.

Specific Data Link Layer Frames:

- The structure of the data link layer frame may be specialized according to the **type of protocol** used i.e. there is **no single fixed format** for a Data Link Layer frame.
- The **fields** and **their sizes** can **change** depending on the **protocol being used**.
- The two different protocols for communication in Data Link Layer are given below:
 1. Point-to-Point Protocol (PPP)
 2. High-level Data Link Control (HDLC)

Point-to-Point Protocol (PPP):

- Point-to-Point Protocol (PPP) is a communication protocol of the data link layer that is used to transmit multiprotocol **data** between **two directly connected** (point-to-point) computers.
- **PPP Use:** Connect **home computers** to the **server** of an **Internet service provider**.
- The fields of a PPP frame are:
 1. **Flag:** It is of **1 byte** that with bit pattern **01111110**.
 2. **Address:** **1 byte** which is set to **11111111** in case of the **broadcast**.
 3. **Control:** **1 byte** set to a constant value of **00000011**.
 4. **Protocol:** **1 or 2 bytes** that define the type of data contained in the **payload field**.
 5. **Payload:** This carries the **data** from the network layer. The **maximum length** of the payload field is **1500 bytes**.
 6. **FCS:** It is a **2 byte or 4 bytes** frame check sequence for **error detection**. The standard code used is **CRC** (cyclic redundancy code).

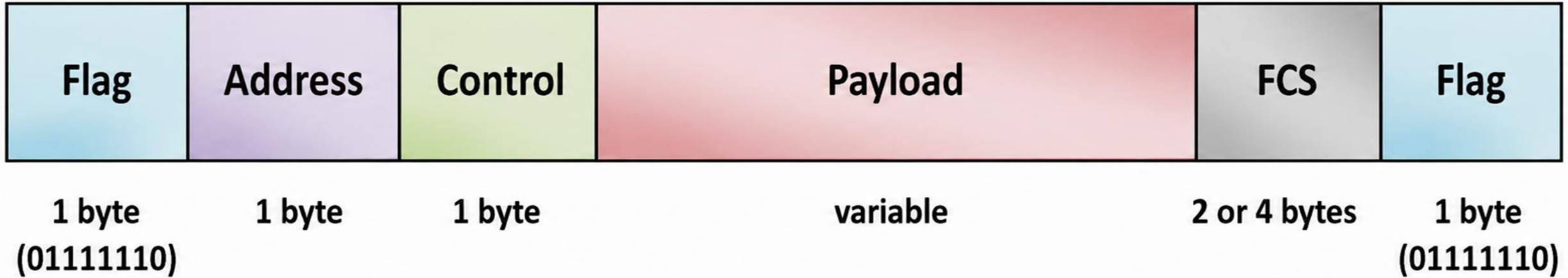
PPP Frame



High-level Data Link Control (HDLC):

- High-level Data Link Control (HDLC) is a group of communication protocols of the data link layer for **transmitting data** between **network points or nodes**.
- The fields of an HDLC frame are:
 1. **Flag:** It is an **8-bit sequence** with bit pattern **01111110**.
 2. **Address:** It contains the **address** of the **receiver**. The address field may be from **1 byte to several bytes** depending on the need of the network.
 3. **Control:** It is **1 or 2 bytes** containing **flow** and **error control** information.
 4. **Payload:** This carries the **data** from the network layer. **Its length** may **vary** from **one network to another**.
 5. **FCS:** It is a **2 byte or 4 bytes** frame check sequence for **error detection**. The standard code used is **CRC** (cyclic redundancy code).

HDLC Frame



Flow Control:

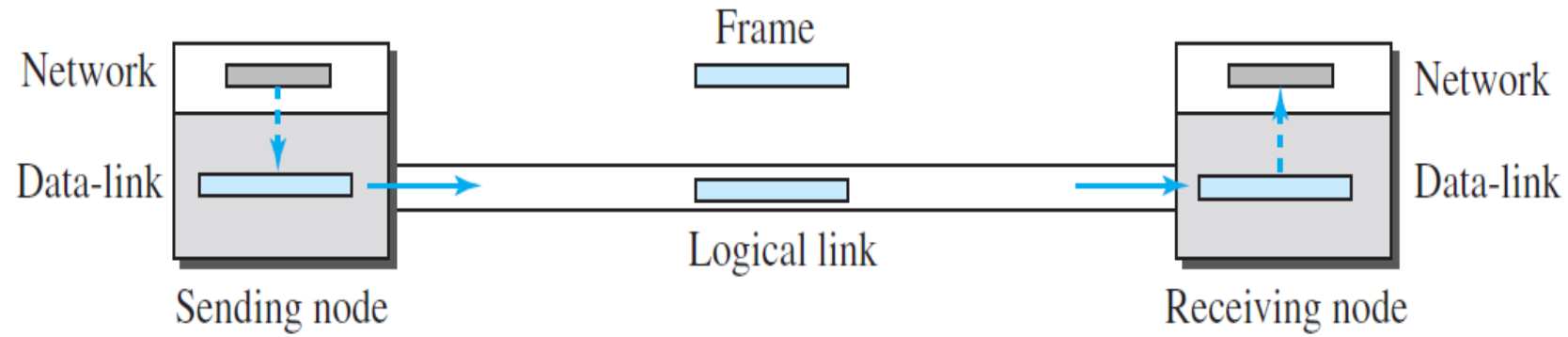
- Flow control is a technique that **allows two stations** working at **different speeds** to **communicate** with each other.
- It is a set of measures taken to regulate the **amount of data** that a sender sends so that a **fast sender** does **not overwhelm** a **slow receiver**.
- In data link layer, the sender continues to **send frames** only after it has **received acknowledgments** from the user for the **previous frames**. This is called **feedback-based flow control**.
- Here, a **restriction** is imposed on the **number of frames** the sender can **send before** it **waits** for an **acknowledgment** from the **receiver**.

Feedback based Flow Control Techniques:

- Data link layer uses **feedback-based flow control** mechanisms.
- There are two main techniques –
 1. Simple Protocol
 2. Stop and Wait Protocol
 3. Sliding Window Protocol

Simple Protocol:

- Our first protocol is a **simple protocol** with **neither flow nor error control**.
- We assume that the receiver can **immediately handle any frame it receives**. In other words, the receiver can **never** be **overwhelmed** with **incoming frames**.
- The data-link layer at the sender **gets a packet** from its network layer, **makes a frame** out of it, and **sends the frame**.
- The data-link layer at the receiver **receives a frame** from the link, **extracts the packet** from the frame, and **delivers the packet** to its **network layer**.
- The data-link layers of the sender and receiver provide **transmission services** for **their network layers**.



Simple Protocol

Stop and Wait:

This protocol involves the following transitions –

- The sender **sends a frame** and **waits for acknowledgment**.
- Once the receiver **receives** the frame, it **sends an acknowledgment** frame back to the sender.
- On receiving the **acknowledgment frame**, the sender understands that the receiver is **ready to accept the next frame**. So, it **sends the next frame in queue**.

Example:

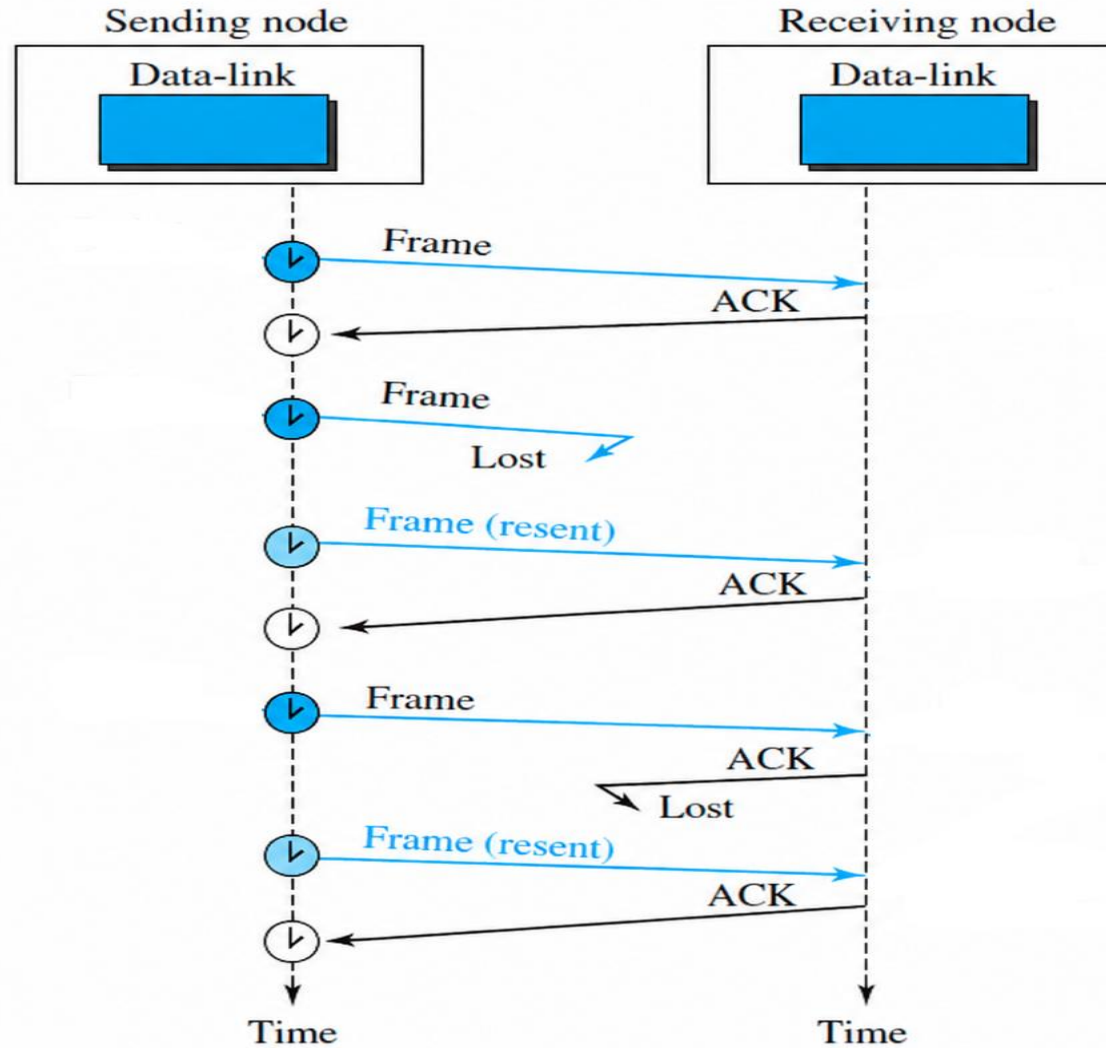
Legend

-  Start the timer.
-  Stop the timer.
-  Restart a time-out timer.

Notes:

A *lost* frame means either lost or corrupted.

A *lost* ACK means either lost or corrupted.



Sliding Window:

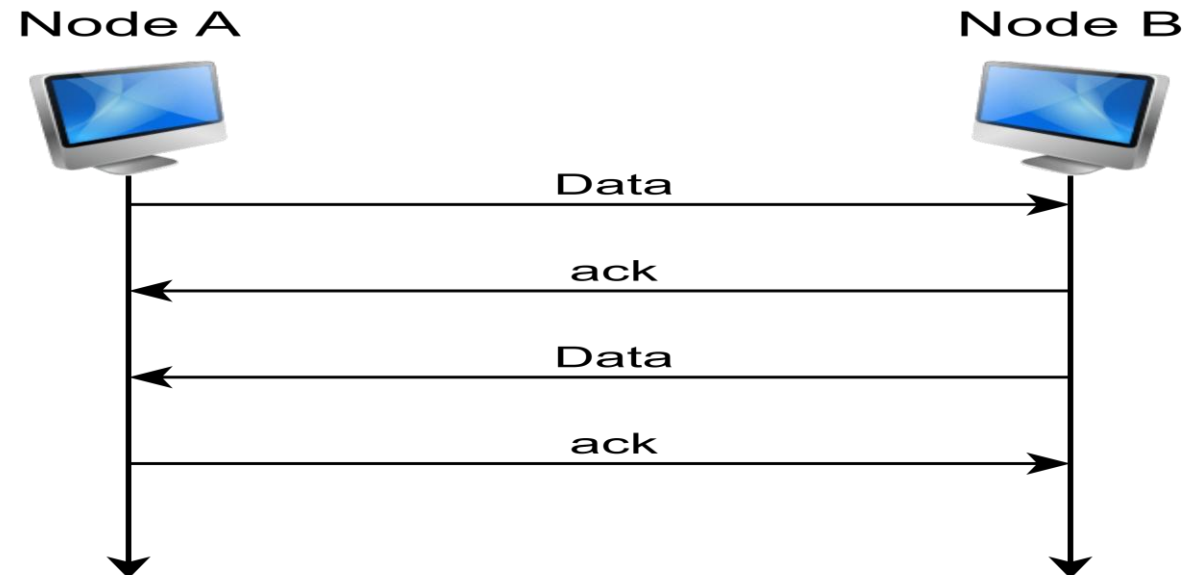
- This protocol **improves** the **efficiency** of **stop and wait protocol** by allowing **multiple frames** to be transmitted **before** receiving an **acknowledgment**.
- The working principle of this protocol can be described as follows –
 1. Both the sender and the receiver has **finite sized buffers** called **windows**.
 2. The sender and the receiver agrees upon the **number of frames to be sent** based upon the **buffer size**.
 3. The sender sends **multiple frames** in a sequence, **without waiting** for **acknowledgment**.
 4. When its **sending window** is **filled**, it **waits** for **acknowledgment**.
 5. On **receiving acknowledgment**, it **advances the window** and transmits the **next frames**, according to the number of **acknowledgments received**.

Piggybacking:

- Piggybacking is a technique used in two-way communication to make data transfer more efficient. Instead of sending a separate acknowledgment (ack) frame, the acknowledgment is included along with the data being sent in the opposite direction.
- Example:** Suppose Node A and Node B are communicating in both directions.

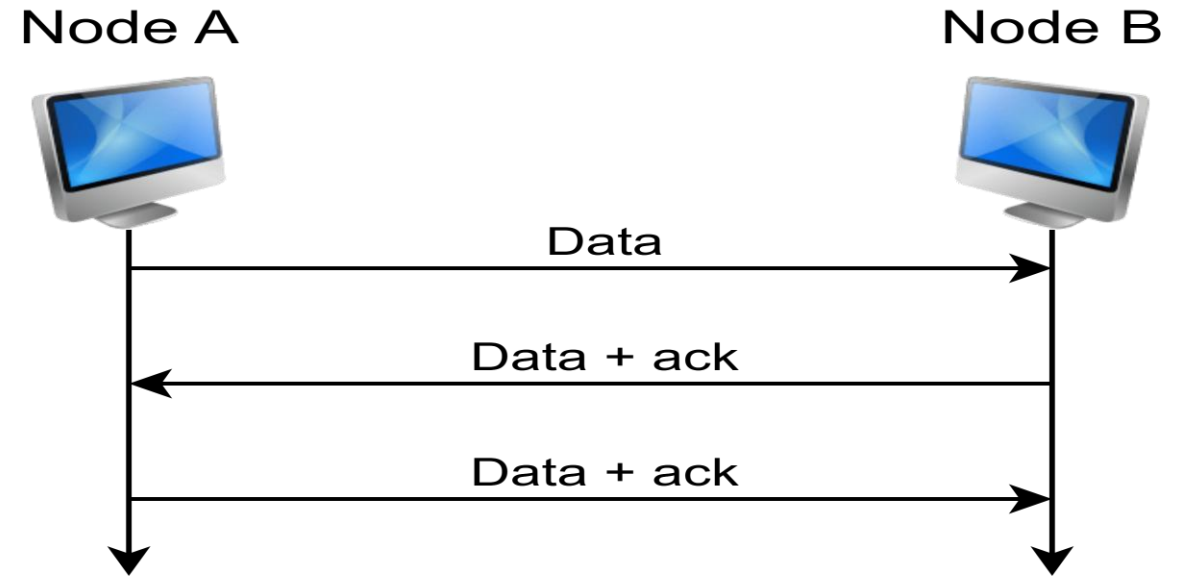
Without Piggybacking:

Here, separate ack frames are sent.



With Piggybacking:

When A sends data to B, B may already have data to send to A. B can include the **ack** for A's data together with B's own data in the same frame. Similarly, A can acknowledge B's data while sending its own data.



Advantages:

- 1. Reduces the number of frames:** The ack does not need to be transmitted as a separate frame.
- 2. Reduces overhead:** A separate ack frame has its own header and other control information. Combining ack with data reduces this overhead.
- 3. Improves network efficiency:** The same transmission can carry both data and acknowledgment.
- 4. Better utilization of the communication link:** The link is used more effectively when both nodes have data to exchange.
- 5. Useful for two-way communication:** It is particularly useful when data is flowing in both directions.



THANK YOU